Project Proposal Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
					Total obtained mark	
Comments						

Semester: Spring 2026

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Batch: 40 **Section: E1**

Course Code: SE331 **Course Name: Software Engineering Design Capstone Project**

Course Teacher Name: Rahat Uddin Azad

Designation: Lecturer

Submission Date: 08.02.2026

TakaTracker

TakaTracker is a simple but powerful tool (often a spreadsheet, app, or lightweight software) that helps very small businesses manage their products, sales, and profits in one place.

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1. Introduction

1.1 Background Overview

Application Domain: Financial Technology (FinTech) / Digital Bookkeeping

Bangladesh has a thriving micro, small, and medium enterprise (MSME) sector with over 7.8 million establishments employing approximately 31 million people. Despite this significant contribution to the economy, most small merchants still rely on traditional paper-based bookkeeping methods (locally known as "khata" or ledger) to track their daily transactions, customer dues, and inventory.

Current Practices:

- **Paper-based ledgers:** Most small shop owners maintain handwritten records in physical notebooks, which are prone to damage, loss, and calculation errors.
- **Memory-based tracking:** Many merchants rely on memory to track customer dues, leading to disputes and revenue loss.
- **Limited financial visibility:** Without proper records, merchants cannot analyze their business performance, identify trends, or make data-driven decisions.
- **Existing digital solutions:** While apps like TallyKhata, Hisabee, DokanKhata, and Sheba exist, many are either too complex for non-tech-savvy users, require constant internet connectivity, or lack proper Bengali language support.

The COVID-19 pandemic accelerated digital adoption, but a significant gap remains in providing accessible, offline-capable financial tools tailored to the Bangladeshi market.

1.2 Problem Statement

Core Problem:

Small merchants in Bangladesh face significant challenges in managing their daily business transactions, tracking customer/supplier dues, and maintaining accurate financial records due to reliance on traditional paper-based bookkeeping methods.

Specific Limitations of Current Systems:

Problem	Impact
Paper ledgers are easily damaged or lost	Complete loss of financial records
Manual calculations lead to errors	Revenue leakage and disputes
No internet connectivity in rural areas	Existing apps become unusable
Complex interfaces intimidate users	Low adoption rates

Limited Bengali language support	Communication barriers
No backup or sync capabilities	Data isolation and vulnerability

Affected Stakeholders:

- **Small merchants** (grocery stores, pharmacies, hardware shops) lose track of ৳10,000-50,000 monthly in uncollected dues.
- **Suppliers** struggle with credit tracking.

Why Software Solution is Needed:

A mobile application with offline-first capability, Bengali language support, and simple UI can bridge this digital divide while preserving the familiar "khata" experience that merchants trust.

1.3 Objectives

General Objective:

To develop a user-friendly, offline-capable mobile application that digitizes traditional bookkeeping practices for Bangladeshi small merchants, enabling them to efficiently track transactions, manage customer dues, and gain insights into their business performance.

Specific Objectives:

1. Design an intuitive UI supporting both Bengali and English languages
2. Implement offline-first architecture allowing full functionality without internet
3. Develop comprehensive transaction management (sales, purchases, expenses, payments)
4. Create customer/supplier management with automatic due calculations
5. Build inventory tracking with low-stock alerts
6. Integrate local payment methods (bKash, Nagad, Rocket) for QR-based payments
7. Implement data synchronization with conflict resolution for multi-device support
8. Generate daily, weekly, and monthly business reports

1.4 Scope

In-Scope Features:

Module	Features
Authentication	Phone number + OTP verification, JWT tokens
Dashboard	Daily summary, receivables/payables, quick actions
Transactions	Sales, purchases, expenses, payments (in/out), returns
Contacts	Customer/supplier management, due tracking, credit limits
Inventory	Product catalog, stock levels, low-stock alerts
Payments	QR code generation, payment method selection
Reports	Daily/weekly/monthly summaries, dues reports
Sync	Offline-first with background synchronization
Localization	Full Bengali and English support

Out-of-Scope Features:

- Advanced accounting features (GST, VAT calculations)
- Multi-branch business management
- Payroll and employee management
- E-commerce integration
- Bank account integration (only payment gateway stubs)
- Real-time inventory tracking with barcode scanning
- iOS deployment (Android-only for this phase)

System Boundaries:

- Target platform: Android (API 21+)
- Backend: REST API (no real-time WebSocket)
- Payment: Stub implementation only (no actual payment processing)
- Storage: Local SQLite with cloud sync (no offline-to-offline sync)

1.5 Stakeholders

Stakeholder	Role	Interest
Small Merchants	Primary end users	Track transactions, reduce errors, collect dues faster
Suppliers	Secondary users	Track credit extended to merchants
Shop Staff	End users (optional)	Record transactions on behalf of owner
System Administrators	Maintain backend	Ensure uptime, manage data, handle support

1.6 Proposed Solution

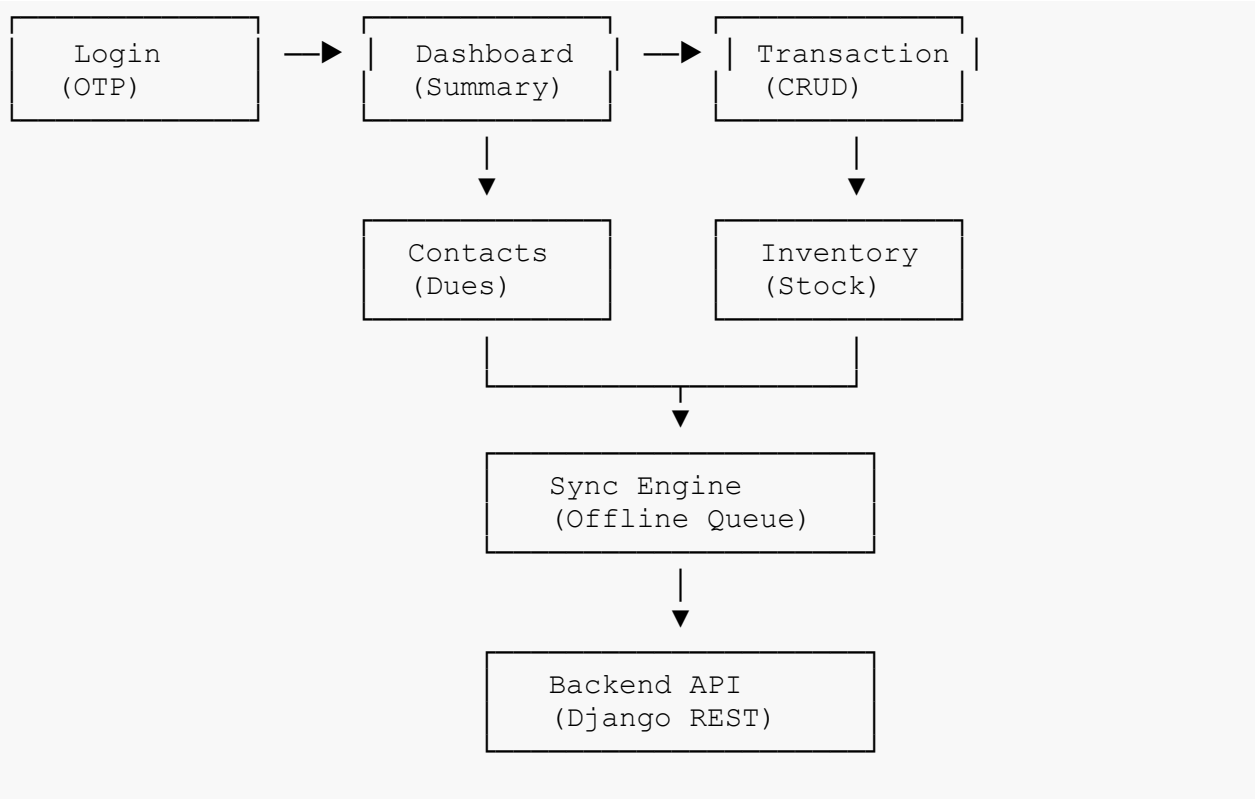
Solution Overview:

A cross-platform mobile application built with React Native (Expo) that provides a digital "khata" experience. The app works completely offline, syncing data when connectivity is available.

Key Features Addressing Problems:

Problem	Solution
Paper damage/loss	Cloud-synced digital storage
Calculation errors	Automatic calculations
No internet	Offline-first with local database
Complex interfaces	Simple, Bengali-first UI
No backups	Automatic cloud sync

System Workflow:



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2. System Requirements

The system shall allow users to register using phone number and OTP verification
The system shall allow users to log in using verified phone credentials
The system shall display a dashboard with today's sales, purchases, and dues summary
The system shall allow users to create sales transactions with customer selection
The system shall allow users to create purchase transactions with supplier selection
The system shall allow users to record expense entries with categories
The system shall allow users to record payment received (from customers)
The system shall allow users to record payment made (to suppliers)
The system shall automatically calculate customer/supplier balance
The system shall allow users to add/edit/delete customers and suppliers
The system shall display list of customers with dues (receivables)
The system shall display list of suppliers with dues (payables)
The system shall allow users to add/edit/delete products
The system shall track product stock levels
The system shall alert users when stock falls below threshold
The system shall generate daily transaction reports
The system shall work completely offline with local data storage
The system shall sync data to server when connectivity is available
The system shall support Bengali and English languages
The system shall generate QR codes for payment collection

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3. Tools and Technologies

3.1 Frontend (Mobile Application)

Category	Technology	Justification
Framework	React Native (Expo)	Cross-platform, large ecosystem, easy deployment
Language	TypeScript	Type safety, better developer experience
State Management	Zustand	Lightweight, simple API, offline persistence
Navigation	React Navigation 6	Industry standard for RN apps
Forms	React Hook Form + Zod	Performant forms with schema validation
Styling	StyleSheet (Vanilla)	Maximum control, no external dependencies
Local Storage	AsyncStorage	Simple key-value storage for preferences
HTTP Client	Axios	Request/response interceptors, timeout handling
Localization	i18n-js	Lightweight internationalization

3.2 Backend (REST API)

Category	Technology	Justification
Framework	Django 5.0	Robust, batteries-included Python framework
API	Django REST Framework	Industry standard for REST APIs
Language	Python 3.11+	Readable, extensive libraries
Database	PostgreSQL 15	Reliable, ACID-compliant, JSON support
Cache	Redis	Session storage, rate limiting
Authentication	JWT (djangorestframework-simplejwt)	Stateless, secure tokens
Containerization	Docker + Docker Compose	Consistent development environment

3.3 Development Tools

Tool	Purpose
VS Code	Primary IDE
Git + GitHub	Version control
Postman	API testing
Expo Go	Mobile app testing
pgAdmin	Database management

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4. Project Timeline and Work Plan

4.1 Milestone-Based Timeline (12 Weeks)

Week	Milestone	Deliverables	Team Member

1-2	**Project Setup**	Proposal, architecture design, environment setup	All
3-4	**Backend Core**	Database models, auth APIs, CRUD endpoints	Sudipto
3-4	**Mobile Setup**	Navigation, theme, i18n, auth screens	Shazid
5-6	**Transaction Module**	Transaction API + mobile screens	All
7-8	**Contact & Inventory**	Contact/product management	All
9-10	**Sync & Offline**	Sync engine, conflict resolution	Masud
9-10	**Reports & QR**	Report screens, QR payment	All
11	**Integration Testing**	End-to-end testing, bug fixes	All
12	**Documentation**	User manual, final submission	All

4.2 Detailed Task Breakdown

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Phase 1: Foundation (Week 1-2)
├── Project proposal and approval
├── Technology selection and setup
├── Git repository initialization
├── Docker development environment
└── Initial database schema design

Phase 2: Backend Development (Week 3-4)
├── User authentication (OTP, JWT)
├── Contact management APIs
├── Transaction management APIs
├── Inventory management APIs
└── Sync endpoint implementation

Phase 3: Mobile Development (Week 3-8)
├── Authentication screens
├── Dashboard with summary
├── Transaction CRUD screens
├── Contact management screens
├── Inventory screens
├── Settings and profile

Phase 4: Advanced Features (Week 9-10)
├── Offline data persistence
├── Background sync logic
├── Conflict resolution
├── Report generation
└── QR payment integration

Phase 5: Testing & Deployment (Week 11-12)
├── Unit testing
├── Integration testing
├── Bug fixes
├── APK build
└── Documentation
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5. Optional Sections

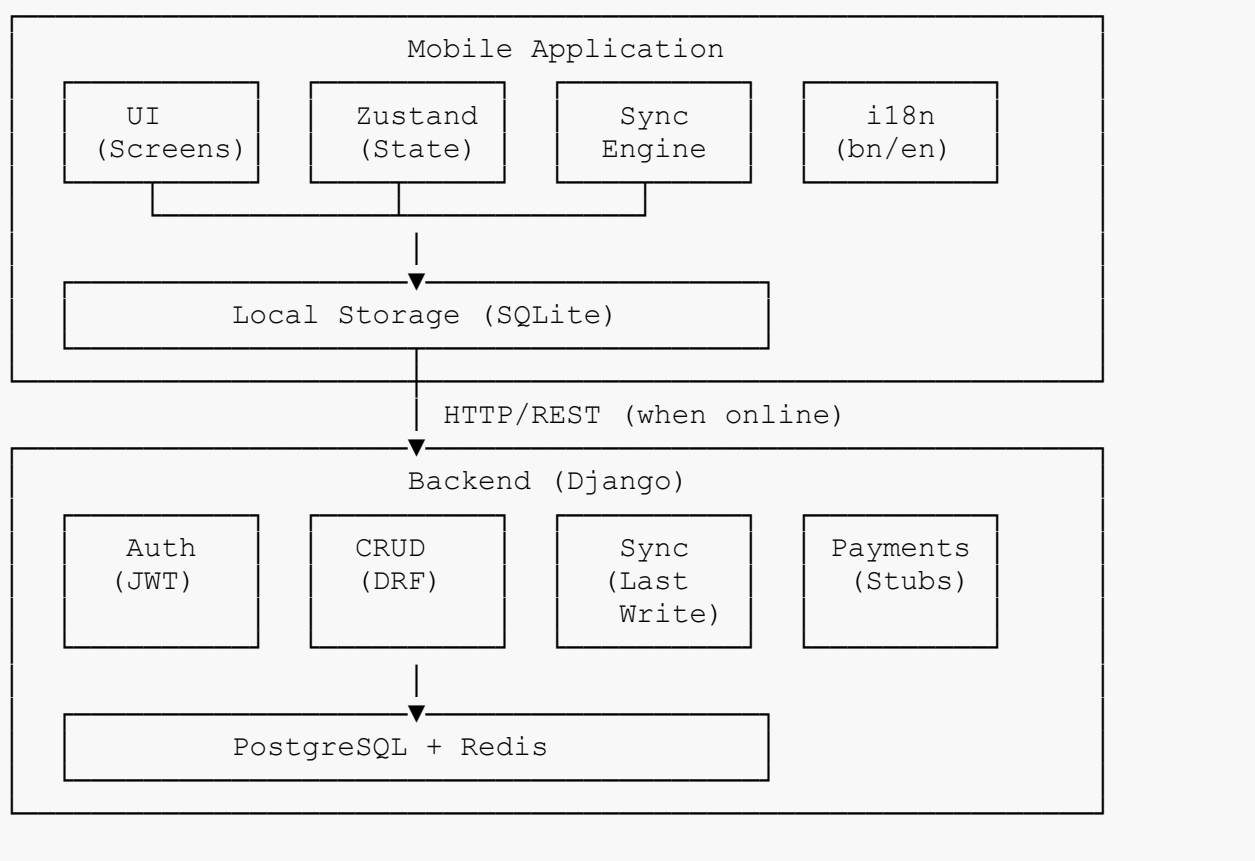
5.1 Existing Systems Analysis

Application	Pros	Cons
TallyKhata	Well-established, feature-rich	Requires internet for many features
DokanKhata	Good UI	Limited offline support
Hishabee	Comprehensive	Complex for basic merchants
Paper Ledger	No tech needed	Easily lost, calculation errors

Our Differentiator:

True offline-first architecture with seamless sync, combined with a simplified Bengali-first interface tailored for basic smartphone users.

5.2 System Architecture



5.3 Ethical, Legal, and Social Considerations

Ethical Considerations:

- **Data Privacy:** User financial data is sensitive. All data is encrypted and users have full control over their data export/deletion.
- **Digital Divide:** The app is designed for users with limited tech literacy, avoiding discrimination based on education level.
- **Accessibility:** UI follows accessibility guidelines with appropriate font sizes and color contrast.

Legal Considerations:

- **Data Protection:** Compliance with Bangladesh's proposed Data Protection Act
- **Payment Regulations:** Payment features are stubs only; actual implementation would require Bangladesh Bank approval

- **Terms of Service:** Clear T&C informing users about data usage

Social Impact:

- Empowers small merchants with better financial management
- Reduces disputes over dues between merchants and customers
- Promotes financial inclusion in underserved communities

5.4 Future Work

After the initial release, the following enhancements are planned:

1. **iOS Version:** Port the application to iOS platform
2. **Barcode Scanner:** Add product barcode scanning for faster entry
3. **Advanced Reports:** Profit/loss statements, cash flow analysis
4. **Multi-branch Support:** For merchants with multiple locations
5. **SMS Reminders:** Automated due payment reminders to customers
6. **WhatsApp Integration:** Share invoices via WhatsApp
7. **Bank Integration:** Direct bank account linking for automatic transaction import
8. **Voice Input:** Add products/transactions via Bengali voice commands

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Conclusion

The TakaTracker Clone project addresses a genuine need in the Bangladeshi small business ecosystem by providing an accessible, offline-capable digital bookkeeping solution. By combining modern mobile development practices with an understanding of local user needs, this application aims to bridge the digital divide and empower millions of small merchants to better manage their businesses.

References

Bangladesh Bureau of Statistics - MSME Survey 2019